

Inverse of a Matrix: A square matrix A is said to be invertible if there exists a unique matrix B such that $AB=BA=I$, where I is the identity matrix. Then B is called the inverse of A and it is denoted by $B=A^{-1}$.

Mathematically,

$$A^{-1} = \frac{\text{adj.}A}{|A|} \quad ; \text{where, } |A| \neq 0$$

Note:1. A matrix A has inverse iff it is non-singular. i.e. $|A| \neq 0$.

2. The inverse of a matrix, if it exists, is unique.

Cofactors of a square matrix: If A be any $n \times n$ square matrix

$$\text{i.e., } A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

The determinant of A is,

$$|A| = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$$

Then the cofactor of its any entry a_{ij} is defined as,

$$A_{ij} = (-1)^{i+j} \left(\begin{array}{l} \text{determinant of submatrix made by} \\ \text{deleting } i\text{th row and } j\text{th column of } A \end{array} \right).$$

Adjoint matrix: Let $A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$ be any $n \times n$ square matrix and A_{ij} be the

cofactors of entries a_{ij} , then the matrix of cofactors from A is,

$$\begin{pmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \dots & \dots & \dots & \dots \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{pmatrix}$$

The transpose of this matrix is called the adjoint of $\mathbf{A}$ and is denoted by $\text{adj}(\mathbf{A})$.

$$\text{i.e., } \text{adj}(\mathbf{A}) = \begin{pmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \dots & \dots & \dots & \dots \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{pmatrix}^t = \begin{pmatrix} A_{11} & A_{21} & \dots & A_{n1} \\ A_{12} & A_{22} & \dots & A_{n2} \\ \dots & \dots & \dots & \dots \\ A_{1n} & A_{2n} & \dots & A_{nn} \end{pmatrix}.$$

Problem-01: Find the adjoint matrix of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix}$.

Solution: The given matrix is,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix}$$

The cofactors of each elements of $|A|$ are,

$$\begin{aligned} A_{11} &= \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} & A_{12} &= -\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} & A_{13} &= \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} & A_{21} &= -\begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} & A_{22} &= \begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} & A_{23} &= -\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} \\ &= 6-1 & &= -(4-3) & &= 2-9 & &= -(4-3) & &= 2-9 & &= -(1-6) \\ &= 5 & &= -1 & &= -7 & &= -1 & &= -7 & &= 5 \end{aligned}$$

$$\begin{aligned} A_{31} &= \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} & A_{32} &= -\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} & A_{33} &= \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} \\ &= 2-9 & &= -(1-6) & &= 3-4 \\ &= -7 & &= 5 & &= -1 \end{aligned}$$

The matrix of cofactors is,

$$\begin{pmatrix} 5 & -1 & -7 \\ -1 & -7 & 5 \\ -7 & 5 & -1 \end{pmatrix}$$

$$\therefore \text{adj}.A = \begin{pmatrix} 5 & -1 & -7 \\ -1 & -7 & 5 \\ -7 & 5 & -1 \end{pmatrix}^t$$

$$= \begin{pmatrix} 5 & -1 & -7 \\ -1 & -7 & 5 \\ -7 & 5 & -1 \end{pmatrix}$$

This is required adjoint matrix.

Problem-02: Find the adjoint matrix of the matrix $A = \begin{pmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{pmatrix}$.

Solution: The given matrix is,

$$A = \begin{pmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{pmatrix}$$

The cofactors of each elements of $|A|$ are,

$$A_{11} = -4 ; A_{12} = 1 ; A_{13} = 4 ; A_{21} = -3 ; A_{22} = 0 ; A_{23} = 4 ; A_{31} = -3 ; A_{32} = 1 ; A_{33} = 3.$$

The matrix of cofactors is,

$$\begin{pmatrix} -4 & 1 & 4 \\ -3 & 0 & 4 \\ -3 & 1 & 3 \end{pmatrix}$$

$$\therefore \text{adj}.A = \begin{pmatrix} -4 & 1 & 4 \\ -3 & 0 & 4 \\ -3 & 1 & 3 \end{pmatrix}^t$$

$$= \begin{pmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{pmatrix}$$

This is required adjoint matrix.

Problem-03: Find the inverse of the matrix $A = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}$.

Solution: The given matrix is,

$$A = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}$$

The determinant of A is,

$$\begin{aligned} |A| &= \begin{vmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{vmatrix} \\ &= 2(12-2) - 3(16-1) + 4(8-3) \\ &= 20 - 45 + 20 \\ &= -5 \end{aligned}$$

Since, $|A| \neq 0$. So the given matrix is a non-singular matrix and it has an inverse matrix.

The cofactors of each elements of $|A|$ are,

$$A_{11} = 10 ; A_{12} = -15 ; A_{13} = 5 ; A_{21} = -4 ; A_{22} = 4 ; A_{23} = -1 ; A_{31} = -9 ; A_{32} = 14 ; A_{33} = -6.$$

The matrix of cofactors is,

$$\begin{pmatrix} 10 & -15 & 5 \\ -4 & 4 & -1 \\ -9 & 14 & -6 \end{pmatrix}$$

$$\begin{aligned}\therefore \text{adj.}A &= \begin{pmatrix} 10 & -15 & 5 \\ -4 & 4 & -1 \\ -9 & 14 & -6 \end{pmatrix}^t \\ &= \begin{pmatrix} 10 & -4 & -9 \\ -15 & 4 & 14 \\ 5 & -1 & -6 \end{pmatrix}\end{aligned}$$

Therefore,

$$\begin{aligned}A^{-1} &= \frac{\text{adj.}A}{|A|} \\ &= -\frac{1}{5} \begin{pmatrix} 10 & -4 & -9 \\ -15 & 4 & 14 \\ 5 & -1 & -6 \end{pmatrix}\end{aligned}$$

This is required inverse matrix.

Problem-04: If $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$ then find AB^{-1} .

Solution: The given matrices are,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$$

The determinant of **B** is,

$$\begin{aligned}|B| &= \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix} \\ &= 1(18-12) - 1(9-3) + 1(4-2) \\ &= 6 - 6 + 2 \\ &= 2\end{aligned}$$

Since, $|B| \neq 0$. So it is a non-singular matrix and it has an inverse matrix.

The cofactors of each elements of $|B|$ are,

$$B_{11}=6 ; B_{12}=-6 ; B_{13}=2; B_{21}=-5 ; B_{22}=8 ; B_{23}=-3; B_{31}=1 ; B_{32}=-2 ; B_{33}=1.$$

The matrix of cofactors is,

$$\begin{pmatrix} 6 & -6 & 2 \\ -5 & 8 & -3 \\ 1 & -2 & 1 \end{pmatrix}$$

$$\therefore \text{adj}.A = \begin{pmatrix} 6 & -6 & 2 \\ -5 & 8 & -3 \\ 1 & -2 & 1 \end{pmatrix}^t$$

$$= \begin{pmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{pmatrix}$$

$$\begin{aligned} \therefore B^{-1} &= \frac{\text{adj}.B}{|B|} \\ &= \frac{1}{2} \begin{pmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{pmatrix} \end{aligned}$$

Therefore,

$$\begin{aligned} AB^{-1} &= \frac{1}{2} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{pmatrix} \begin{pmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 0 & 2 & 0 \\ -2 & 4 & 0 \\ 0 & -1 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 & 0 \\ -1 & 2 & 0 \\ 0 & -\frac{1}{2} & \frac{3}{2} \end{pmatrix} \end{aligned}$$

This is required answer.

Exercise:

1. Find the adjoint matrix of the matrix $A = \begin{pmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{pmatrix}$.

2. Find the adjoint matrix of the matrix $A = \begin{pmatrix} 1 & 0 & 3 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$.

3. Find the inverse of the matrix $A = \begin{pmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{pmatrix}$.

4. Find the inverse of the matrix $A = \begin{pmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{pmatrix}$.

5. If $A = \begin{pmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{pmatrix}$ then find $A^{-1}B$.

6. If $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 1 & 5 & 7 \end{pmatrix}$ then show that $AA^{-1} = I$.

